

Prime Numbers and the Fundamental Theorem of Arithmetic

- **Definition 1**

An integer $p > 1$ is a **prime number** if its only positive divisors are 1 and p . An integer $n > 1$ that is not prime is called **composite**.

- **Example 1.1**

The first prime numbers are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, \dots$$

The numbers

$$4, 6, 8, 9, 10, 12, \dots$$

are composite.

Note that 1 is neither prime nor composite.

- **Proposition 1.2**

An integer $n > 1$ is composite, if and only if, $n = ab$ for some $a, b \in \mathbb{N}$ with $1 < a < n$ and $1 < b < n$.

- **Proof 1.2.1**

($\Rightarrow$) Suppose that n is composite, then it has a positive divisor a with $a \neq 1, n$. Hence there exists $b \in \mathbb{N}$ such that $n = ab$ with $1 < a < n$ and $1 < b < n$, because if $b = 1$ then $a = n$.

($\Leftarrow$) Suppose that $n = ab$ for some $a, b \in \mathbb{N}$ with $1 < a < n$ and $1 < b < n$. Then such a factorization makes a a divisor of n with $a \neq 1, n$. Therefore n is composite.

- **Lemma 1.3**

Let p be a prime and let $a \in \mathbb{Z}$. If $p \nmid a$, then there exist $x, y \in \mathbb{Z}$ such that $ax + py = 1$.

- **Proof 1.3.1**

Consider the set

$$S = \{ax + py : x, y \in \mathbb{Z} \wedge ax + py > 0\}.$$

Since $p \nmid a$, we have that $a \neq 0$. Therefore $|a| \in S$ because $|a| = a \cdot 1 + p \cdot 0 > 0$ if $a > 0$, or $|a| = a \cdot (-1) + p \cdot 0 > 0$ if $a < 0$.

Now, as $S \subseteq \mathbb{N}$ and $S \neq \emptyset$, by the well-ordering axiom S has a first element d . Therefore $d = ax_0 + py_0 > 0$. If we divide a by d , then by the division algorithm we have that

$$a = dq + r \text{ with } 0 \leq r < |d| = d.$$

Hence

$$r = a - dq = a - (ax_0 + py_0)q = a(1 - x_0q) + p(-y_0q).$$

If $r > 0$, then $r \in S$, which contradicts the minimality of d . Therefore we obtain that $r = 0$ and $a = dq$, that is, $d \mid a$. Using the same argument, dividing p by d shows that $d \mid p$. Now, as p is prime we have that $d = 1 \vee d = p$, but as $p \nmid a$ we have that $d = 1$. Therefore we obtain that $ax_0 + py_0 = 1$.

- **Corollary 1.4 (Euclid's Lemma)**

(1) Let p be a prime and let $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a \vee p \mid b$.

- **Proof 1.4.1**

Suppose that $p \nmid a$, then by Lemma 1.3 there exist $x, y \in \mathbb{Z}$ such that $ax + py = 1$, hence

$$abx + pby = b.$$

Since $p \mid ab$, there exists $k \in \mathbb{Z}$ such that $ab = kp$, so

$$b = abx + pby = kpx + pby = (kx + by)p \text{ with } kx + by \in \mathbb{Z}.$$

Therefore $p \mid b$.

(2) Let p be a prime and $a_1, a_2, \dots, a_n \in \mathbb{Z}$. If $p \mid (a_1 a_2 \cdots a_n)$, then $p \mid a_i$ for some $1 \leq i \leq n$. In particular, if all the a_j are prime, then $p = a_i$ for some $1 \leq i \leq n$.

■ **Proof 1.4.2**

By induction.

1. Base step

If $n = 1$ then the result is true.

2. Inductive step

Suppose that the result is true for n .

$$\begin{aligned} p \mid (a_1 a_2 \cdots a_n a_{n+1}) &\implies p \mid (a_1 a_2 \cdots a_n) a_{n+1} \\ &\implies p \mid (a_1 a_2 \cdots a_n) \vee p \mid a_{n+1} \\ &\implies p \mid a_i \text{ for some } 1 \leq i \leq n \vee p \mid a_{n+1} \\ &\implies p \mid a_j \text{ for some } 1 \leq j \leq n+1. \end{aligned}$$

Therefore the result is true for $n+1$.

If all the a_j are prime and $p \mid a_i$, then by definition of a prime we have that $p = a_i$.

- **Theorem 2 (Fundamental theorem of arithmetic)** Every integer $n \geq 2$ can be written in a unique form up to rearrangements as a product

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

where $p_1, p_2, \dots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$. For each i , the notation $\nu_{p_i}(n) = \alpha_i$ denotes the exponent of the prime p_i in the factorization of n .

○ **Proof 2.1**

1. Existence

Consider the set

$$A = \{m \in \mathbb{N} : m \geq 2 \wedge m \text{ cannot be written as a product of primes}\}.$$

If $A \neq \emptyset$, then by the well-ordering axiom A has a first element n . Then n is not a prime. As n is composite, there exist $a, b \in \mathbb{N}$ such that $n = ab$ with $1 < a < n$ and $1 < b < n$. As n is the first element of A , we have that a and b can be written as a product of primes, and therefore n can be written as a product of primes, which is a contradiction because $n \in A$. Therefore $A = \emptyset$ and thus every integer ≥ 2 can be written as a product of prime numbers.

2. Uniqueness

Suppose that the result is false, that is, there exists an integer ≥ 2 that can be written in two different forms as a product of primes. Then the set

$$B = \{m \in \mathbb{N} : m \geq 2 \wedge m \text{ can be written in two different forms as a product of primes}\}$$

is nonempty, then by the well-ordering axiom B has a first element n , then

$$p_1 p_2 \cdots p_r = n = q_1 q_2 \cdots q_s.$$

Since $p_1 \mid q_1 q_2 \cdots q_s$, then $p_1 \mid q_i$ for some $1 \leq i \leq s$ by Corollary 1.4(2), and by commutativity of the product we may assume that $i = 1$ and so $p_1 = q_1$. Hence by the cancellation law we have

$$p_2 p_3 \cdots p_r = q_2 q_3 \cdots q_s < n.$$

Therefore $p_2 p_3 \cdots p_r$ and $q_2 q_3 \cdots q_s$ are two different factorizations of a number $< n$, which contradicts the minimality of n . Therefore $B = \emptyset$, that is, every integer ≥ 2 can be written in a unique form as a product of primes.

○ **Example 2.2**

Find the prime factorization of 224 and 1260.

$$224 = 2^5 \cdot 7 \quad 1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7.$$

◦ **Corollary 2.3**

Every integer $n \leq -2$ can be written in a unique form up to rearrangements as a product

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

where $p_1, p_2, \dots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$.

▪ **Proof 2.3.1**

Since $n \leq -2$ then $-n \geq 2$, and by the fundamental theorem of arithmetic

$$-n = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

where $p_1, p_2, \dots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$.

Hence

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}.$$

◦ **Proposition 2.4**

Suppose $x, y \in \mathbb{N}$, then $\nu_p(xy) = \nu_p(x) + \nu_p(y)$ for any prime p .

▪ **Proof 2.4.1**

Consider the prime factorizations of x and y . Let

$$x = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r} \quad \alpha_k \in \mathbb{N}_0,$$

$$y = p_1^{\beta_1}p_2^{\beta_2}\cdots p_r^{\beta_r} \quad \beta_k \in \mathbb{N}_0.$$

Then we have

$$xy = p_1^{\alpha_1+\beta_1}p_2^{\alpha_2+\beta_2}\cdots p_r^{\alpha_r+\beta_r}.$$

By the uniqueness of the prime factorization of xy , for each $p = p_k$, we have that

$$\nu_p(xy) = \alpha_k + \beta_k = \nu_p(x) + \nu_p(y).$$

• **Theorem 3 (Euclid's theorem)**

The set of prime numbers is infinite.

◦ **Proof 3.1**

Suppose that there are only finitely many primes $p_1, p_2, \dots, p_n$ and consider

$$N = p_1p_2\cdots p_n + 1.$$

Since $N > 1$, by the fundamental theorem of arithmetic we have a prime divisor q of N . Then $q = p_i$ for some $1 \leq i \leq n$, hence q divides both N and $p_1p_2\cdots p_n$, and therefore q divides $N - p_1p_2\cdots p_n = 1$, which is impossible because q is a prime. Therefore there are infinitely many primes.